

Database Design Project

Oracle Baseball League Store Database

Project Scenario:

You are a small consulting company specializing in database development. You have just been awarded the contract to develop a data model for a database application system for a small retail store called Oracle Baseball League (OBL).

The Oracle Baseball League store serves the entire surrounding community selling baseball kit. The OBL has two types of customer, there are individuals who purchase items like balls, cleats, gloves, shirts, screen printed t-shirts, and shorts. Additionally customers can represent a team when they purchase uniforms and equipment on behalf of the team.

Teams and individual customers are free to purchase any item from the inventory list, but teams get a discount on the list price depending on the number of players. When a customer places an order we record the order items for that order in our database.

OBL has a team of three sales representatives that officially only call on teams but have been known to handle individual customer complaints.

Section 4 Lesson 1 Exercise : Oracle SQL Developer Data Modeler

Creating a Logical Data Model (S4L1 Objective 1)

Oracle SQL Developer Data Modeler allows you to create Entities, attributes, and UIDs with correct optionality and cardinality, Supertype and subtype entities as well as Arcs, hierarchical, barred, and recursive relationships.

1. The first step in building a logical model (ERD) in Oracle SQL Developer Data Modeler is to click the Logical tab.
 - a. If you do not see the Logical tab, then perform the following steps:
 - b. Right-click the Logical Model in the browser.
 - c. Select Show.
2. Create entities.
3. Create UIDs for each entity.
 - a. Click the entity for which you want to define the UID.
 - b. With Attributes selected in the left navigator of the Entity Properties window, select the attribute that you want to assign as the UID.
 - c. Select the Primary UID check box.
4. Create mandatory or optional attributes for each entity (at this stage don't apply any data types to the attributes). Add comments to explain the attribute where necessary. Do not include any foreign key attributes as these are added through the relationships.
5. Create the relationship between the entities. You can create M:N Relation (many-to-many), 1:N Relation (one-to-many), 1:N Relation identifying (one-to-many barred) or 1:1 Relation (one-to-one).
 - a. Set the source entity and target entity for the relationship.
 - b. Name the relationship following the convention of naming the entity on the one side followed by a colon and naming the second entity in the relationship (customer : customer address).
 - c. Use name on source and name on target to add the labels to the relationships. These won't be displayed on the logical model but will be shown on the physical model.
6. A subtype entity inherits the properties of the supertype.
 - a. To define an entity as a subtype in Oracle SQL Developer Data Modeler you need to ensure that the supertype exists.
 - b. When you create the entity choose the super drop down and identify the supertype entity.
7. Save your work.

Suggested task starting point

Recreate your ERD in Oracle SQL Developer Data Modeler.

